

Classified as hazardous according to criteria of EPA New Zealand

## Section 1 - Identification

**Product Name** tert-Butyl 4-(2-formylphenyl)piperazine-1-carboxylate

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | CC30704CB; CC30704DA; CC30704DE; CC30704ZZ                                                  |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:NZinfo@thermofisher.com">NZinfo@thermofisher.com</a>                        |

**Recommended Use** Laboratory chemicals.

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

- 6.1D - Substances that are acutely toxic (Oral)
- 6.1D - Substances that are acutely toxic (Dermal)
- 6.3A - Substances that are irritating to the skin
- 6.4A - Substances that are irritating to the eye
- 6.1D - Substances that are acutely toxic (Inhalation)
- 6.1E - Substances that are acutely toxic (Inhalation)

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### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                                    |            |
|----------------------------------------------------|------------|
| Acute Oral Toxicity                                | Category 4 |
| Acute Dermal Toxicity                              | Category 4 |
| Acute Inhalation Toxicity - Dusts and Mists        | Category 4 |
| Skin Corrosion/Irritation                          | Category 2 |
| Serious Eye Damage/Eye Irritation                  | Category 2 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements



**Signal Word**

**Warning**

**Hazard Statements**

H302 - Harmful if swallowed  
H312 - Harmful in contact with skin  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H332 - Harmful if inhaled  
H335 - May cause respiratory irritation

**Precautionary Statements**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor/physician if you feel unwell  
P330 - Rinse mouth  
P362 - Take off contaminated clothing and wash before reuse  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

No information available

## Section 3 - Composition and Information on Ingredients

| Component       | CAS-No      | Weight % |
|-----------------|-------------|----------|
| N-acetylleucine | 174855-57-3 | 100      |

## Section 4 - First Aid Measures

|                       |                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>     | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>      | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Skin Contact</b>   | If skin irritation persists, call a physician. Wash off immediately with plenty of water for at least 15 minutes. |
| <b>Eye Contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>General Advice</b> | If symptoms persist, call a physician.                                                                            |

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|                                     |                                                |
|-------------------------------------|------------------------------------------------|
| Self-Protection of the First Aider  | Use personal protective equipment as required. |
| First Aid Facilities                | Eyewash, safety shower and washroom.           |
| Most important symptoms and effects | None reasonably foreseeable.                   |
| Notes to Physician                  | Treat symptomatically.                         |

## **Section 5 - Fire Fighting Measures**

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### **Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **Section 6 - Accidental Release Measures**

### **Emergency procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

### **Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

### **Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **Section 7 - Handling and Storage**

### **Precautions for Safe Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

### **Conditions for Safe Storage, Including any Incompatibilities**

Keep container tightly closed in a dry and well-ventilated place. Keep out of the reach of children. Keep in properly labeled containers.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Exposure limits

The product does not contain any hazardous materials with occupational exposure limits established.

### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material     | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|--------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Disposable gloves. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Skin and body protection

Long sleeved clothing

#### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

### Environmental exposure controls

Do not allow material to contaminate ground water system.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

#### Appearance Physical State

Light yellow  
Solid

#### Odor Odor Threshold pH

No information available  
No data available  
No data available

#### Melting Point/Range

89.2 - 90.8 °C / 192.6 - 195.4 °F

#### Softening Point

No data available

#### Boiling Point/Range

No data available

#### Flash Point

No data available

Method - No information available

|                          |                          |
|--------------------------|--------------------------|
| Evaporation Rate         | No data available        |
| Flammability (solid,gas) | No information available |
| Explosion Limits         | No data available        |

|                                         |                          |             |
|-----------------------------------------|--------------------------|-------------|
| Vapor Pressure                          | No data available        |             |
| Vapor Density                           | No data available        | (Air = 1.0) |
| Specific Gravity / Density              | No data available        |             |
| Bulk Density                            | No data available        |             |
| Water Solubility                        | No information available |             |
| Solubility in other solvents            | No information available |             |
| Partition Coefficient (n-octanol/water) |                          |             |
| Autoignition Temperature                | No data available        |             |
| Decomposition Temperature               | No data available        |             |
| Viscosity                               | No data available        |             |
| Explosive Properties                    | No information available |             |
| Oxidizing Properties                    | No information available |             |

**Other information**

|                   |               |
|-------------------|---------------|
| Molecular Formula | C16 H22 N2 O3 |
| Molecular Weight  | 290.3608      |

## Section 10 - Stability and Reactivity

|            |                                            |
|------------|--------------------------------------------|
| Reactivity | None known, based on information available |
|------------|--------------------------------------------|

|           |                                 |
|-----------|---------------------------------|
| Stability | Stable under normal conditions. |
|-----------|---------------------------------|

|                     |                        |
|---------------------|------------------------|
| Conditions to Avoid | Incompatible products. |
|---------------------|------------------------|

|                        |                          |
|------------------------|--------------------------|
| Incompatible Materials | Strong oxidizing agents. |
|------------------------|--------------------------|

**Hazardous Decomposition Products** Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
|--------------------------|---------------------------|

## Section 11 - Toxicological Information

### Information on Toxicological Effects

|                     |                                                             |
|---------------------|-------------------------------------------------------------|
| Product Information | No acute toxicity information is available for this product |
|---------------------|-------------------------------------------------------------|

|                     |                   |
|---------------------|-------------------|
| (a) acute toxicity; |                   |
| Oral                | No data available |
| Dermal              | No data available |
| Inhalation          | No data available |

|                                |                   |
|--------------------------------|-------------------|
| (b) skin corrosion/irritation; | No data available |
|--------------------------------|-------------------|

|                                    |                   |
|------------------------------------|-------------------|
| (c) serious eye damage/irritation; | No data available |
|------------------------------------|-------------------|

|                                        |                   |
|----------------------------------------|-------------------|
| (d) respiratory or skin sensitization; |                   |
| Respiratory                            | No data available |
| Skin                                   | No data available |

|                             |                   |
|-----------------------------|-------------------|
| (e) germ cell mutagenicity; | No data available |
|-----------------------------|-------------------|

|                                            |                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------|
| (f) carcinogenicity;                       | No data available<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;                 | No data available                                                              |
| (h) STOT-single exposure;                  | No data available                                                              |
| Results / Target organs                    | Respiratory system                                                             |
| (i) STOT-repeated exposure;                | No data available                                                              |
| Target Organs                              | No information available.                                                      |
| (j) aspiration hazard;                     | No data available                                                              |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                 |
| Symptoms / effects, both acute and delayed | No information available                                                       |

## **Section 12 - Ecological Information**

|                                 |                                                                           |
|---------------------------------|---------------------------------------------------------------------------|
| Ecotoxicity effects             | Do not empty into drains.                                                 |
| Persistence and Degradability   | No information available                                                  |
| Bioaccumulative Potential       | No information available                                                  |
| Mobility                        | No information available.                                                 |
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance            |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance            |

## **Section 13 - Disposal Considerations**

|                                     |                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| Contaminated Packaging              | Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.                                                                                                                                                                                                           |
| Other Information                   | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Waste codes should be assigned by the user based on the application for which the product was used.                                                                                            |

## **Section 14 - Transport Information**

### **IMDG/IMO**

|                      |                              |
|----------------------|------------------------------|
| UN-No                | UN2811                       |
| Proper Shipping Name | Toxic solid, organic, n.o.s. |
| Hazard Class         | 6.1                          |
| Packing Group        | III                          |

NZS 5433:2012

|                      |                              |
|----------------------|------------------------------|
| UN-No                | UN2811                       |
| Proper Shipping Name | Toxic solid, organic, n.o.s. |
| Hazard Class         | 6.1                          |
| Packing Group        | III                          |

IATA

|                      |                               |
|----------------------|-------------------------------|
| UN-No                | UN2811                        |
| Proper Shipping Name | TOXIC SOLID, ORGANIC, N.O.S.* |
| Hazard Class         | 6.1                           |
| Packing Group        | III                           |

Environmental hazards No hazards identified

Special Precautions No special precautions required

Additional information None known

## Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

Prohibition or notification/licensing requirements Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

## Section 16 - Other Information

This safety data sheet complies with the requirements of WorkSafe New Zealand Regulations

Legend

AICS - Australian Inventory of Chemical Substances  
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory  
DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List  
IECSC - Chinese Inventory of Existing Chemical Substances  
PICCS - Philippines Inventory of Chemicals and Chemical Substances  
TWA - Time Weighted Average  
IARC - International Agency for Research on Cancer  
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association  
MARPOL - International Convention for the Prevention of Pollution from Ships  
NZS 5433:2012 - Transport of Dangerous Goods on Land  
LD50 - Lethal Dose 50%  
EC50 - Effective Concentration 50%

NZIoC - New Zealand Inventory of Chemicals  
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
ENCS - Japanese Existing and New Chemical Substances  
KECL - Korean Existing and Evaluated Chemical Substances  
CAS - Chemical Abstracts Service  
ACGIH - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)  
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code  
ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail  
OECD - Organisation for Economic Co-operation and Development  
LC50 - Lethal Concentration 50%  
ATE - Acute Toxicity Estimate

**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
VOC (volatile organic compound)

**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

**Key literature references and sources for data**

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Revision Date** 11-Aug-2020  
**Revision Summary** Initial Release

**Disclaimer**

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**End of Safety Data Sheet**